

PCBC H Complex Numbers & Trig Form: Memorize Sheet

everything below is from memory on quiz day (non-calculator)

Basics

$$i = \sqrt{-1}, i^2 = -1$$

$$\sqrt{-a} = i\sqrt{a} \text{ before multiplying: } \sqrt{-a}\sqrt{-b} = -\sqrt{ab}$$

standard (rectangular) form: $a + bi$

powers: $i, -1, -i, 1$ repeat; exponent $\div 4$, use the remainder

$$\frac{1}{i} = -i$$

Classify

every number is *complex*

real: $b = 0$

pure imaginary: $a = 0, b \neq 0$

nonreal complex: $b \neq 0$

list every one that applies

Arithmetic

add/subtract: real with real, imaginary with imaginary

multiply: FOIL, then $i^2 = -1$

conjugate of $a + bi$ is $a - bi$; $(a + bi)(a - bi) = a^2 + b^2$

divide: multiply top and bottom by the conjugate of the bottom

Quadratics

set = 0 first

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$b^2 - 4ac < 0$: two nonreal conjugate answers

$x^2 = -k$: $x = \pm i\sqrt{k}$ (don't drop $\pm$)

check a solution: plug in, it must equal 0

Complex plane

$a + bi$ graphs as the vector from $(0, 0)$ to (a, b)

horizontal: real axis; vertical: imaginary axis

sum = resultant = diagonal of the parallelogram

$$|a + bi| = \sqrt{a^2 + b^2} \text{ (modulus = length)}$$

Trig (polar) form

$$r(\cos \theta + i \sin \theta) = r \text{ cis } \theta, \theta \in [0^\circ, 360^\circ)$$

$$x = r \cos \theta, y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}, \tan \theta = \frac{y}{x}$$

sketch first, then pick the quadrant

$$|r \text{ cis } \theta| = r$$

Quadrant fix (ref = reference angle)

QI: $\theta = \text{ref}$ QII: $180^\circ - \text{ref}$

QIII: $180^\circ + \text{ref}$ QIV: $360^\circ - \text{ref}$

on an axis: +real 0° , +imag 90° ,

−real 180° , −imag 270°

radians: multiply by $\frac{180^\circ}{\pi}$

Exact values

θ	$\cos \theta$	$\sin \theta$
30°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$
60°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$

signs by quadrant: QI all +, QII sin +, QIII tan +, QIV cos +

Julia set

$$z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$$

find $| \quad |$ of each result

any $|\text{result}| > 2$: NOT in the set

exactly 2 does not exceed 2: keep going

Product and quotient

$$(r_1 \text{ cis } \theta_1)(r_2 \text{ cis } \theta_2) = r_1 r_2 \text{ cis }(\theta_1 + \theta_2)$$

$$\frac{r_1 \text{ cis } \theta_1}{r_2 \text{ cis } \theta_2} = \frac{r_1}{r_2} \text{ cis }(\theta_1 - \theta_2)$$

Non-Calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

You have 30 minutes.

Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $-\sqrt{-49}$

b) $6 - 2i$

Find the product. Write your answer in standard form. (2 points)

2. $\sqrt{-8} \cdot \sqrt{-18}$

Simplify each power of i . (1 point each)

3.

a) i^{47}

b) $\frac{1}{i^{14}}$

Find the sum or difference. Write your answer in standard form. (2 points)

4. $(6 - 5i) - (-2 + 3i) + (4 - i)$

Find the product. Write your answer in standard form. (2 points)

5. $(3 - 4i)(-2 + 5i)$

Find the quotient. Write your answer in standard form. (3 points)

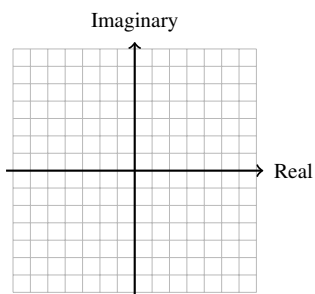
6. $\frac{4 + 7i}{2 - 3i}$

Solve the equation over the set of complex numbers. (4 points)

7. $2x^2 + 3 = -4x$

Find the sum of the pair of complex numbers. Write your answer in standard form. Then graph both complex numbers and their resultant. (3 points)

8. $5 - 2i$, $-3 + 6i$



Write the complex number in rectangular form. (3 points)

9. $8(\cos 300^\circ + i \sin 300^\circ)$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $-3 - 3i$

b) -5

The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = i$ in the Julia set? Show work.

YES / NO

ANSWER KEY – Non-Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $-\sqrt{-49}$

b) $6 - 2i$

a) $-\sqrt{-49} = -7i$; $a = 0$, $b \neq 0$.

b) $a \neq 0$, $b = -2 \neq 0$.

complex, pure imaginary, nonreal complex

complex, nonreal complex

Find the product. Write your answer in standard form. (2 points)

2. $\sqrt{-8} \cdot \sqrt{-18}$

Pull out i first: $\sqrt{-8} = 2i\sqrt{2}$, $\sqrt{-18} = 3i\sqrt{2}$

$(2i\sqrt{2})(3i\sqrt{2}) = 6i^2(2) = 12i^2 = \boxed{-12}$ ($\sqrt{144} = 12$ is the trap: the sign flips.)

Simplify each power of i . (1 point each)

3.

a) i^{47}

b) $\frac{1}{i^{14}}$

a) $47 \div 4 = 11 \text{ R } 3$, so $i^{47} = i^3 = \boxed{-i}$

b) $14 \div 4 = 3 \text{ R } 2$, so $i^{14} = i^2 = -1$; $\frac{1}{-1} = \boxed{-1}$

Find the sum or difference. Write your answer in standard form. (2 points)

4. $(6 - 5i) - (-2 + 3i) + (4 - i)$

$6 - 5i + 2 - 3i + 4 - i$ real: $6 + 2 + 4 = 12$ imaginary: $-5 - 3 - 1 = -9$

$\boxed{12 - 9i}$

Find the product. Write your answer in standard form. (2 points)

5. $(3 - 4i)(-2 + 5i)$

$= -6 + 15i + 8i - 20i^2 = -6 + 23i + 20 = \boxed{14 + 23i}$

Find the quotient. Write your answer in standard form. (3 points)

6. $\frac{4 + 7i}{2 - 3i}$

Multiply by the conjugate of the bottom: $\frac{4 + 7i}{2 - 3i} \cdot \frac{2 + 3i}{2 + 3i}$

top: $8 + 12i + 14i + 21i^2 = -13 + 26i$ bottom: $4 + 9 = 13$

$\frac{-13 + 26i}{13} = \boxed{-1 + 2i}$

Solve the equation over the set of complex numbers. (4 points)

7. $2x^2 + 3 = -4x$

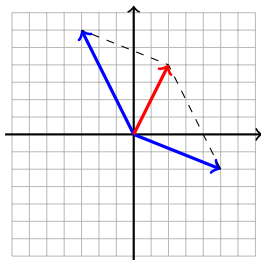
$2x^2 + 4x + 3 = 0$, $a = 2$, $b = 4$, $c = 3$ $b^2 - 4ac = 16 - 24 = -8$

$$x = \frac{-4 \pm \sqrt{-8}}{4} = \frac{-4 \pm 2i\sqrt{2}}{4} = \boxed{-1 \pm \frac{\sqrt{2}}{2}i}$$

Find the sum of the pair of complex numbers. Write your answer in standard form. Then graph both complex numbers and their resultant. (3 points)

8. $5 - 2i$, $-3 + 6i$

$(5 - 3) + (-2 + 6)i = \boxed{2 + 4i}$



Resultant (red) is the diagonal of the parallelogram.

Write the complex number in rectangular form. (3 points)

9. $8(\cos 300^\circ + i \sin 300^\circ)$

300° : QIV, reference 60° ; $\cos 300^\circ = \frac{1}{2}$, $\sin 300^\circ = -\frac{\sqrt{3}}{2}$

$8\left(\frac{1}{2}\right) + 8\left(-\frac{\sqrt{3}}{2}\right)i = \boxed{4 - 4i\sqrt{3}}$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $-3 - 3i$

b) -5

a) QIII. $r = \sqrt{9+9} = 3\sqrt{2}$

$\tan \theta = \frac{-3}{-3} = 1$, ref 45°

$\theta = 180^\circ + 45^\circ = 225^\circ$

$\boxed{3\sqrt{2}(\cos 225^\circ + i \sin 225^\circ)}$

b) negative real axis

$r = 5$, $\theta = 180^\circ$

$\boxed{5(\cos 180^\circ + i \sin 180^\circ)}$

The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = i$ in the Julia set? Show work.

$i^2 - 1 = -2$, $|-2| = 2$: exactly 2 does not exceed 2, keep going.

$(-2)^2 - 1 = 3$, $|3| = 3 > 2$.

NO

Non-Calculator

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You have 30 minutes.

Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $-\sqrt{25}$

b) $2i\sqrt{3}$

Find the quotient. Write your answer in standard form. (2 points)

2. $\frac{\sqrt{-54}}{\sqrt{6}}$

Simplify each power of i . (1 point each)

3.

a) i^{30}

b) $\frac{1}{i^9}$

Is the given number a solution of the equation? Show work. (2 points)

4. $3 - 2i$; $x^2 - 6x + 13 = 0$

YES / NO

Find the product. Write your answer in standard form. (2 points)

5. $(2 - 7i)^2$

Find the quotient. Write your answer in standard form. (3 points)

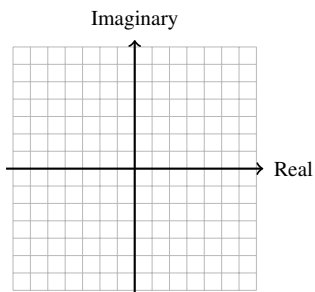
6. $\frac{5 - i}{3 + 2i}$

Solve the equation over the set of complex numbers. (4 points)

7. $3x^2 + 5 = 2x$

Find the sum of the pair of complex numbers. Write your answer in standard form. Graph both complex numbers and their resultant. Then find the absolute value of the sum. (3 points)

8. $-4 + i$, $1 - 5i$



Write the complex number in rectangular form. (3 points)

9. $4 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $1 - i\sqrt{3}$

b) $-7i$

The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = \frac{1}{2}i$ in the Julia set? Show work.

YES / NO

ANSWER KEY – Non-Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $-\sqrt{25}$

b) $2i\sqrt{3}$

a) -5 ; $b = 0$.

b) $a = 0$, $b = 2\sqrt{3} \neq 0$.

complex, real

complex, pure imaginary, nonreal complex

Find the quotient. Write your answer in standard form. (2 points)

2. $\frac{\sqrt{-54}}{\sqrt{6}}$

$\sqrt{-54} = i\sqrt{54} = 3i\sqrt{6}$ $\frac{3i\sqrt{6}}{\sqrt{6}} = \boxed{3i}$

Simplify each power of i . (1 point each)

3.

a) i^{30}

b) $\frac{1}{i^9}$

a) $30 \div 4 = 7 \text{ R } 2$, so $i^{30} = i^2 = \boxed{-1}$

b) $i^9 = i$; $\frac{1}{i} \cdot \frac{-i}{-i} = \frac{-i}{1} = \boxed{-i}$

Is the given number a solution of the equation? Show work. (2 points)

4. $3 - 2i$; $x^2 - 6x + 13 = 0$

$(3 - 2i)^2 = 9 - 12i + 4i^2 = 5 - 12i$ $-6(3 - 2i) = -18 + 12i$

$(5 - 12i) + (-18 + 12i) + 13 = 0 \checkmark$

YES

Find the product. Write your answer in standard form. (2 points)

5. $(2 - 7i)^2$

$(2 - 7i)(2 - 7i) = 4 - 14i - 14i + 49i^2 = 4 - 28i - 49 = \boxed{-45 - 28i}$

Squaring each part separately loses the $-28i$.

Find the quotient. Write your answer in standard form. (3 points)

6. $\frac{5 - i}{3 + 2i}$

$\frac{5 - i}{3 + 2i} \cdot \frac{3 - 2i}{3 - 2i}$ top: $15 - 10i - 3i + 2i^2 = 13 - 13i$ bottom: $9 + 4 = 13$

$\frac{13 - 13i}{13} = \boxed{1 - i}$

Solve the equation over the set of complex numbers. (4 points)

7. $3x^2 + 5 = 2x$

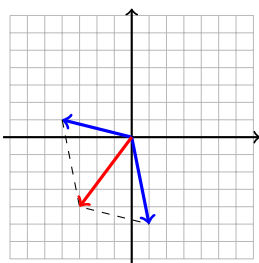
$$3x^2 - 2x + 5 = 0 \quad b^2 - 4ac = 4 - 60 = -56 \quad \sqrt{-56} = 2i\sqrt{14}$$

$$x = \frac{2 \pm 2i\sqrt{14}}{6} = \boxed{\frac{1}{3} \pm \frac{\sqrt{14}}{3}i}$$

Find the sum of the pair of complex numbers. Write your answer in standard form. Graph both complex numbers and their resultant. Then find the absolute value of the sum. (3 points)

8. $-4 + i, 1 - 5i$

$$(-4 + 1) + (1 - 5)i = \boxed{-3 - 4i} \quad |-3 - 4i| = \sqrt{9 + 16} = \boxed{5}$$



Write the complex number in rectangular form. (3 points)

9. $4 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$

$$\frac{5\pi}{6} = 150^\circ: \text{QII, reference } 30^\circ; \cos = -\frac{\sqrt{3}}{2}, \sin = \frac{1}{2}$$

$$4 \left(-\frac{\sqrt{3}}{2} \right) + 4 \left(\frac{1}{2} \right) i = \boxed{-2\sqrt{3} + 2i}$$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $1 - i\sqrt{3}$

b) $-7i$

a) QIV. $r = \sqrt{1 + 3} = 2$

$\tan \theta = -\sqrt{3}$, ref 60°

$\theta = 360^\circ - 60^\circ = 300^\circ$

$$\boxed{2(\cos 300^\circ + i \sin 300^\circ)}$$

b) negative imaginary axis

$r = 7, \theta = 270^\circ$

$$\boxed{7(\cos 270^\circ + i \sin 270^\circ)}$$

The Julia set: compute $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$. If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = \frac{1}{2}i$ in the Julia set? Show work.

$$z^2 - 1 = -\frac{1}{4} - 1 = -\frac{5}{4}, \quad \left| -\frac{5}{4} \right| \leq 2 \quad \left(-\frac{5}{4} \right)^2 - 1 = \frac{9}{16} \leq 2 \quad \left(\frac{9}{16} \right)^2 - 1 = -\frac{175}{256}, \quad |\cdot| < 1$$

No result exceeds 2.

YES

Non-Calculator

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You have 30 minutes.

Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $7 + \sqrt{-4}$

b) i^2

Find the product. Write your answer in standard form. (2 points)

2. $\sqrt{-20} \cdot \sqrt{-45}$

Simplify each power of i . (1 point each)

3.

a) i^{61}

b) $\frac{1}{i^7}$

Find the sum or difference. Write your answer in standard form. (2 points)

4. $(-3 + 8i) + (5 - 2i) - (-1 + 6i)$

Find the product. Write your answer in standard form. (2 points)

5. $(6 - i)(-2 - 3i)$

Find the quotient. Write your answer in standard form. (3 points)

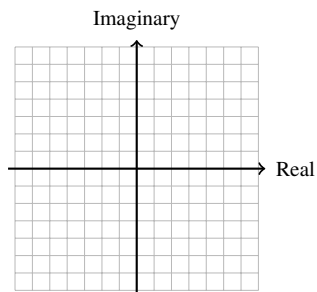
6. $\frac{3 + 11i}{1 + 2i}$

Solve the equation over the set of complex numbers. (4 points)

7. $x^2 + 29 = -4x$

Find the sum of the pair of complex numbers. Write your answer in standard form. Graph both complex numbers and their resultant. Then find the absolute value of the sum. (3 points)

8. $3 + 5i, -6 - i$



Write the complex number in rectangular form. (3 points)

9. $10(\cos 240^\circ + i \sin 240^\circ)$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $2\sqrt{3} - 2i$

b) 6

The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = i\sqrt{2}$ in the Julia set? Show work.

YES / NO

ANSWER KEY – Non-Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $7 + \sqrt{-4}$

b) i^2

a) $7 + 2i$; $b \neq 0$.

b) $i^2 = -1$; $b = 0$.

complex, nonreal complex

complex, real

Find the product. Write your answer in standard form. (2 points)

2. $\sqrt{-20} \cdot \sqrt{-45}$

$\sqrt{-20} = 2i\sqrt{5}$, $\sqrt{-45} = 3i\sqrt{5}$ $(2i\sqrt{5})(3i\sqrt{5}) = 6i^2(5) = 30i^2 =$ -30

Simplify each power of i . (1 point each)

3.

a) i^{61}

b) $\frac{1}{i^7}$

a) $61 \div 4 = 15 \text{ R } 1$, so $i^{61} =$ i

b) $i^7 = i^3 = -i$; $\frac{1}{-i} \cdot \frac{i}{i} = \frac{i}{-i^2} =$ i

Find the sum or difference. Write your answer in standard form. (2 points)

4. $(-3 + 8i) + (5 - 2i) - (-1 + 6i)$

$-3 + 8i + 5 - 2i + 1 - 6i$ real: 3 imaginary: $8 - 2 - 6 = 0$ 3 (or $3 + 0i$)

Find the product. Write your answer in standard form. (2 points)

5. $(6 - i)(-2 - 3i)$

$= -12 - 18i + 2i + 3i^2 = -12 - 16i - 3 =$ $-15 - 16i$

Find the quotient. Write your answer in standard form. (3 points)

6. $\frac{3 + 11i}{1 + 2i}$

$\frac{3 + 11i}{1 + 2i} \cdot \frac{1 - 2i}{1 - 2i}$ top: $3 - 6i + 11i - 22i^2 = 25 + 5i$ bottom: $1 + 4 = 5$

$\frac{25 + 5i}{5} =$ $5 + i$

Solve the equation over the set of complex numbers. (4 points)

7. $x^2 + 29 = -4x$

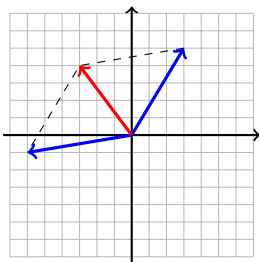
$$x^2 + 4x + 29 = 0 \quad b^2 - 4ac = 16 - 116 = -100$$

$$x = \frac{-4 \pm \sqrt{-100}}{2} = \frac{-4 \pm 10i}{2} = \boxed{-2 \pm 5i}$$

Find the sum of the pair of complex numbers. Write your answer in standard form. Graph both complex numbers and their resultant. Then find the absolute value of the sum. (3 points)

8. $3 + 5i, -6 - i$

$$(3 - 6) + (5 - 1)i = \boxed{-3 + 4i} \quad |-3 + 4i| = \sqrt{9 + 16} = \boxed{5}$$



Write the complex number in rectangular form. (3 points)

9. $10(\cos 240^\circ + i \sin 240^\circ)$

$$240^\circ: \text{QIII, reference } 60^\circ; \cos 240^\circ = -\frac{1}{2}, \sin 240^\circ = -\frac{\sqrt{3}}{2}$$

$$10\left(-\frac{1}{2}\right) + 10\left(-\frac{\sqrt{3}}{2}\right)i = \boxed{-5 - 5i\sqrt{3}}$$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $2\sqrt{3} - 2i$

b) 6

a) QIV. $r = \sqrt{12 + 4} = 4$
 $\tan \theta = \frac{-2}{2\sqrt{3}} = -\frac{1}{\sqrt{3}}, \text{ ref } 30^\circ$
 $\theta = 360^\circ - 30^\circ = 330^\circ$

b) positive real axis
 $r = 6, \theta = 0^\circ$

$$\boxed{4(\cos 330^\circ + i \sin 330^\circ)}$$

$$\boxed{6(\cos 0^\circ + i \sin 0^\circ)}$$

The Julia set: compute $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$. If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = i\sqrt{2}$ in the Julia set? Show work.

$$(i\sqrt{2})^2 - 1 = 2i^2 - 1 = -2 - 1 = -3 \quad |-3| = 3 > 2$$

NO